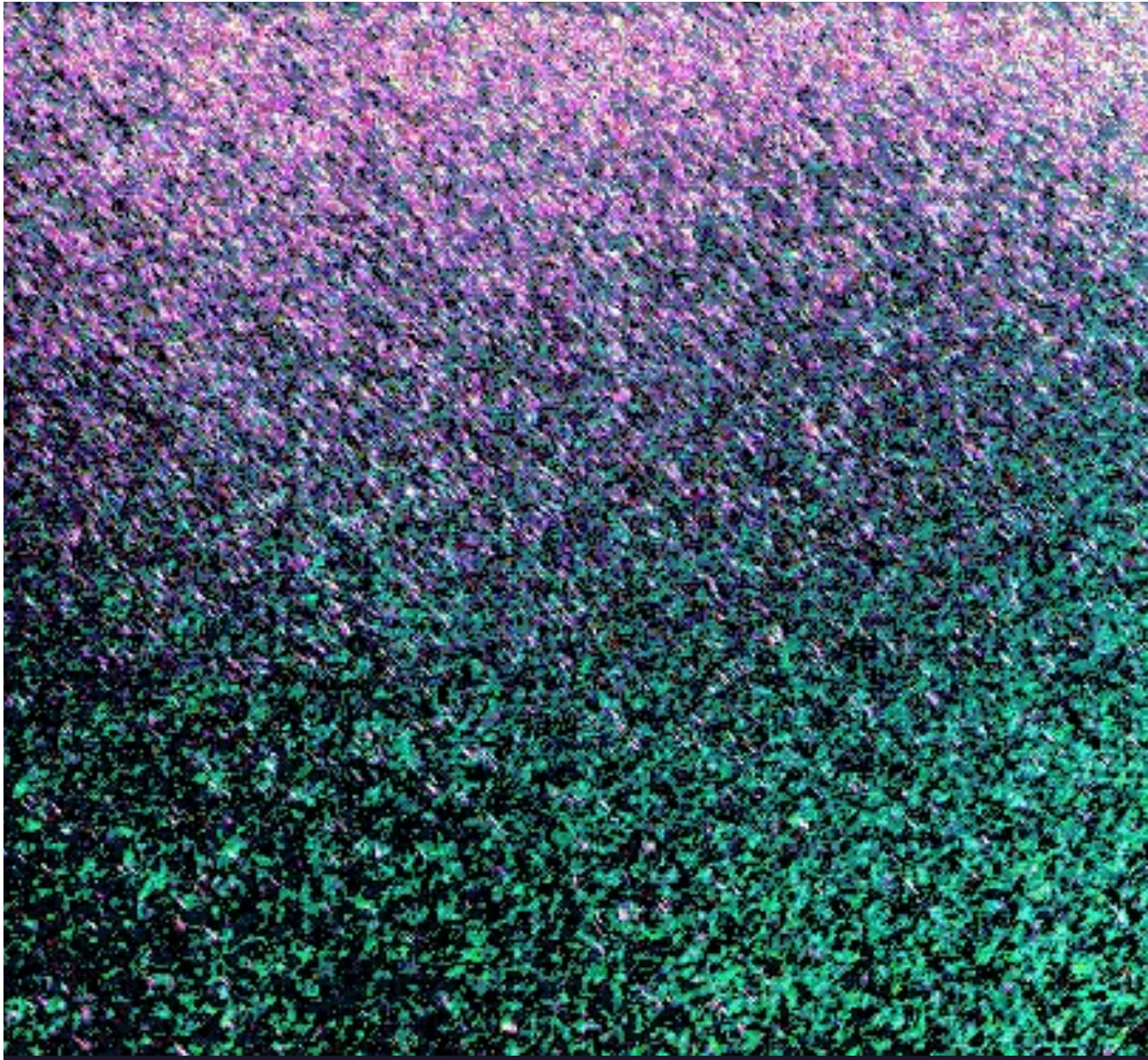
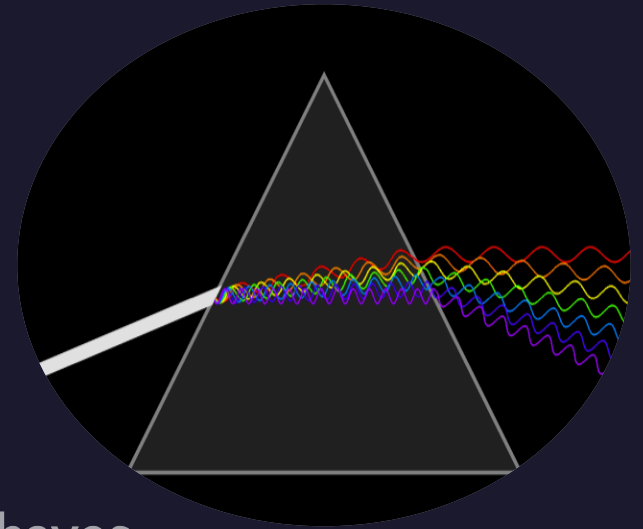




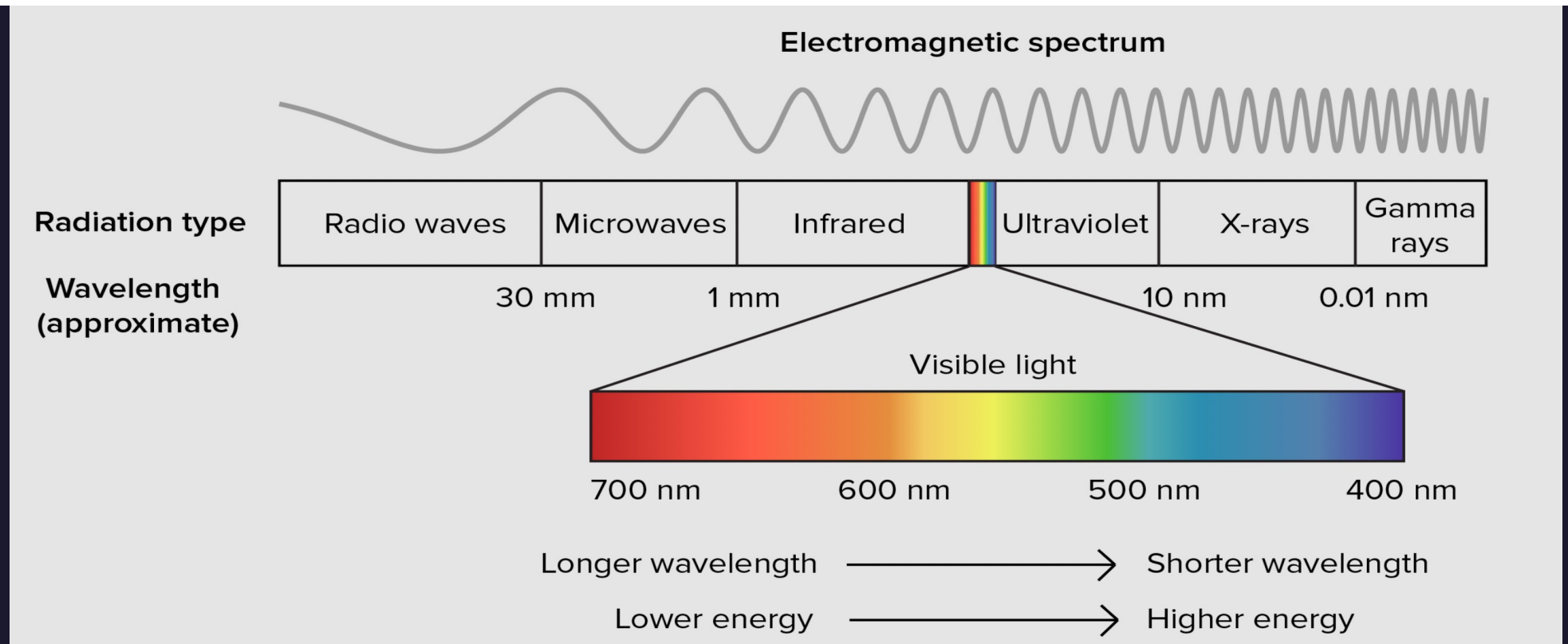
The Electromagnetic Spectrum



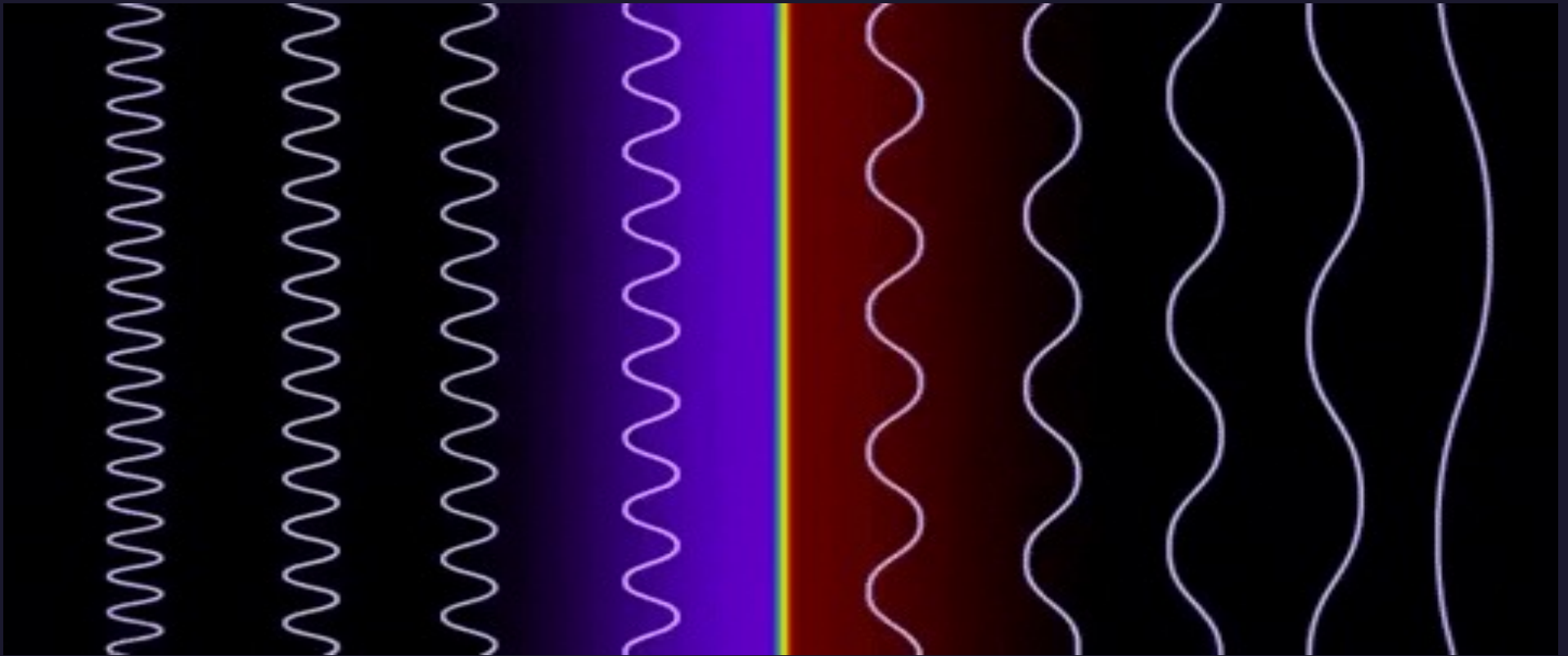
I can...



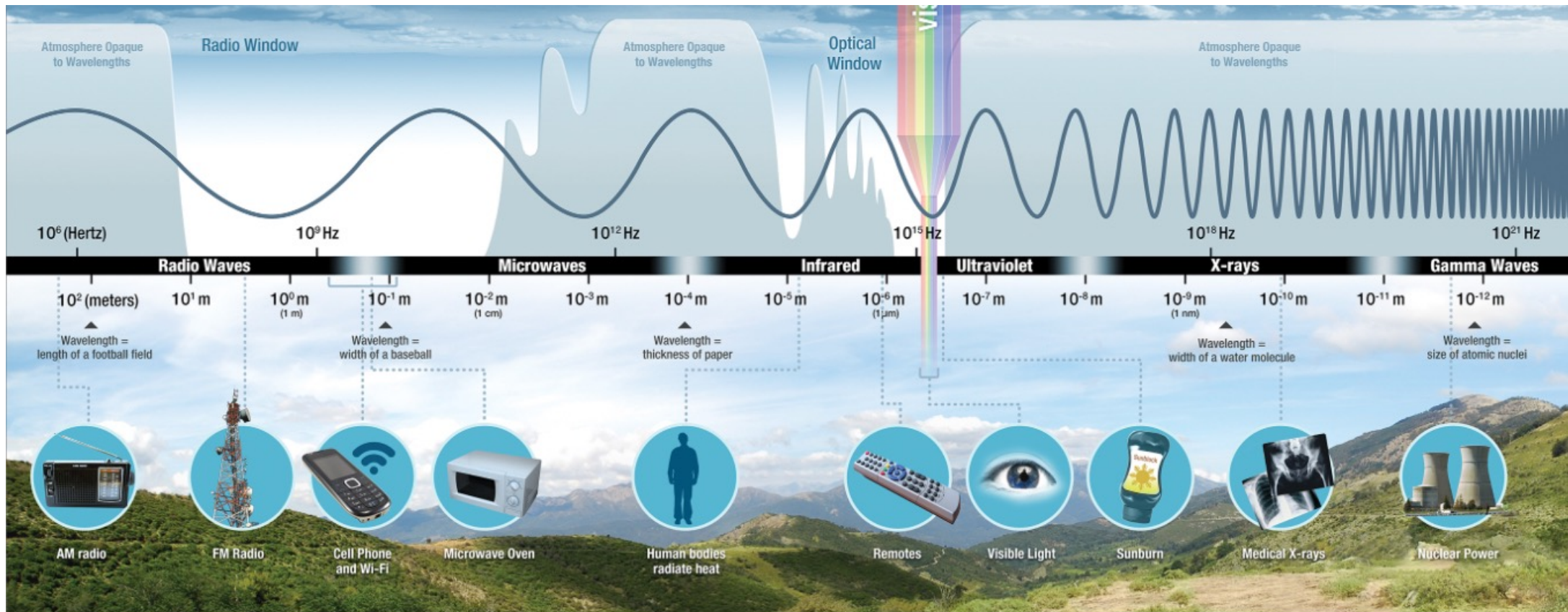
- Define what light is and tell where it comes from
- Explain how the properties of light affect the way it behaves
- Describe how light moves
- Tell how changing mediums affects light
- Tell how light can be transmitted, absorbed, and reflected
- Describe how radiant energy is transformed into other forms of energy



The light we see is a small part of the electromagnetic radiation all around us. Our eyes and brains evolved to make use of this radiation, sight, as a way to keep us alive.



The electromagnetic spectrum shows all the different wavelengths of light (electromagnetic radiation). These wavelengths carry different amounts of energy, so they have different uses for humans.



What are some uses of different energy bands on the electromagnetic spectrum?



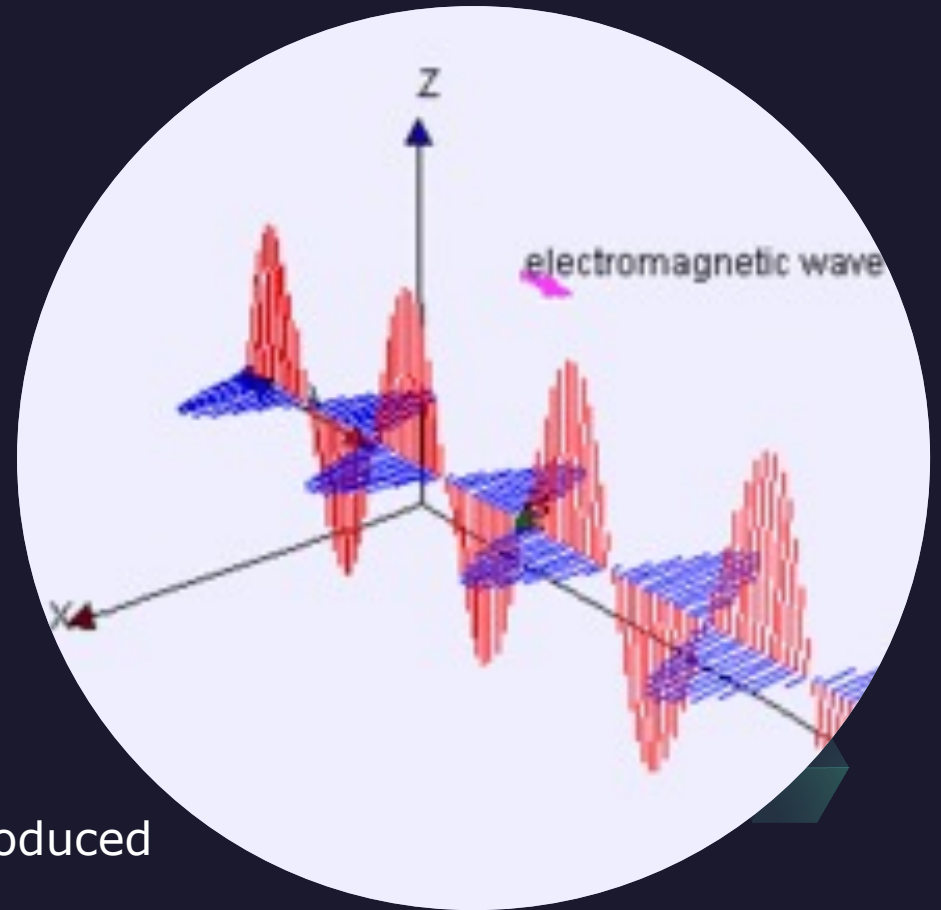
**shorter
wavelength = higher frequency = more energy**

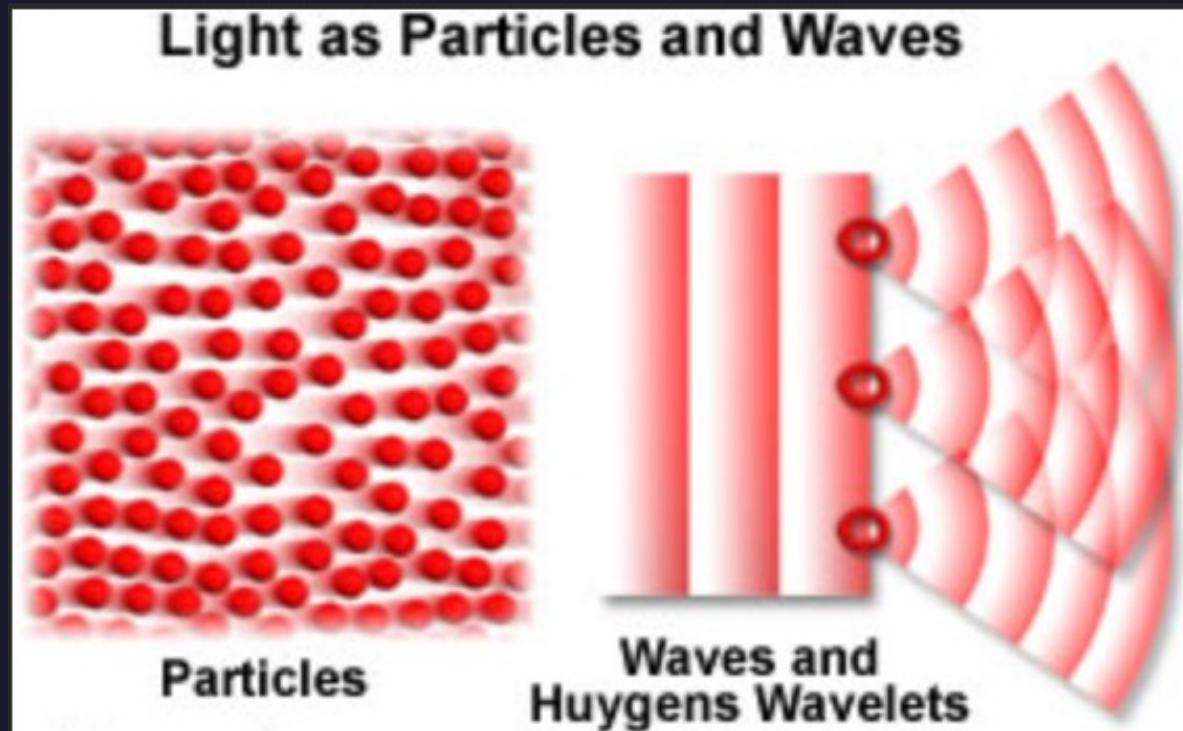
In general, light is made when electrons move, which happens all the time!

What sources of light can you think of?

If heat is the original source of energy, the light produced is called **incandescent light**.

If the original energy source is not heat, then the light produced is called **luminescent light**, also called “cold light.”

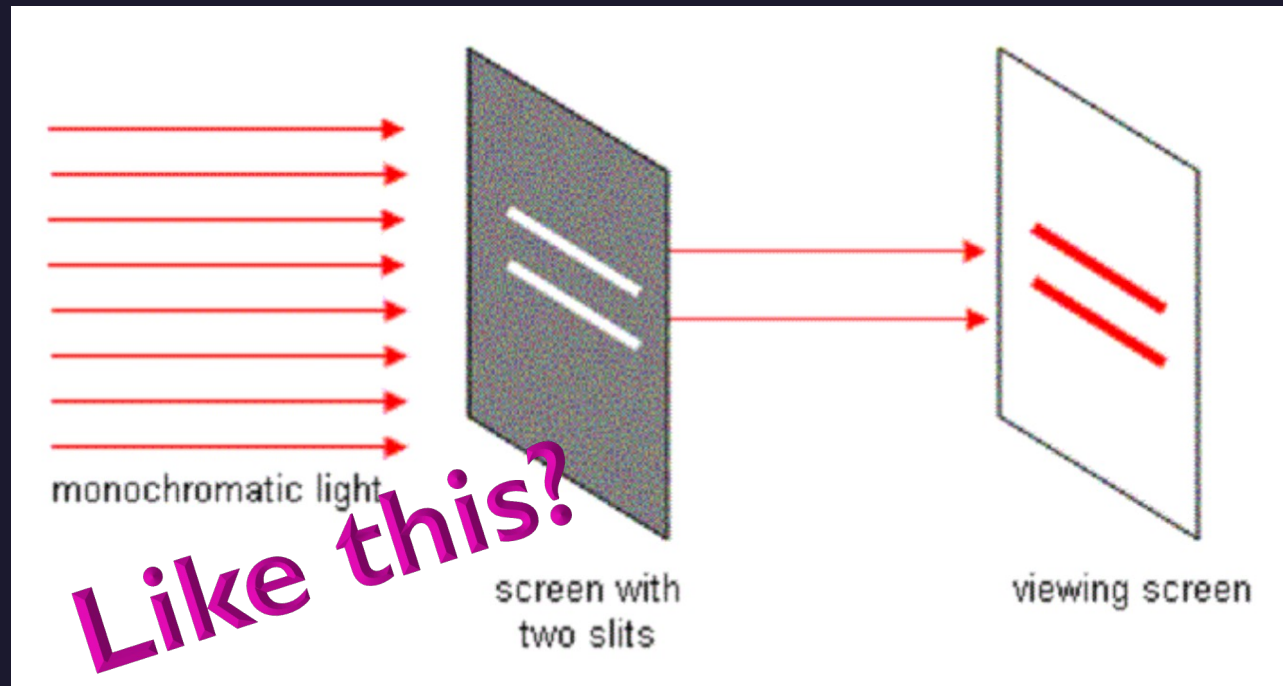




Light has the properties of both a **particle** and a **wave**. This is called **wave-particle duality**

Light as a **wave**

Imagine you cut two tiny cuts in a card and let the light shine through onto another card in a dark room. What would the light on the card look like?

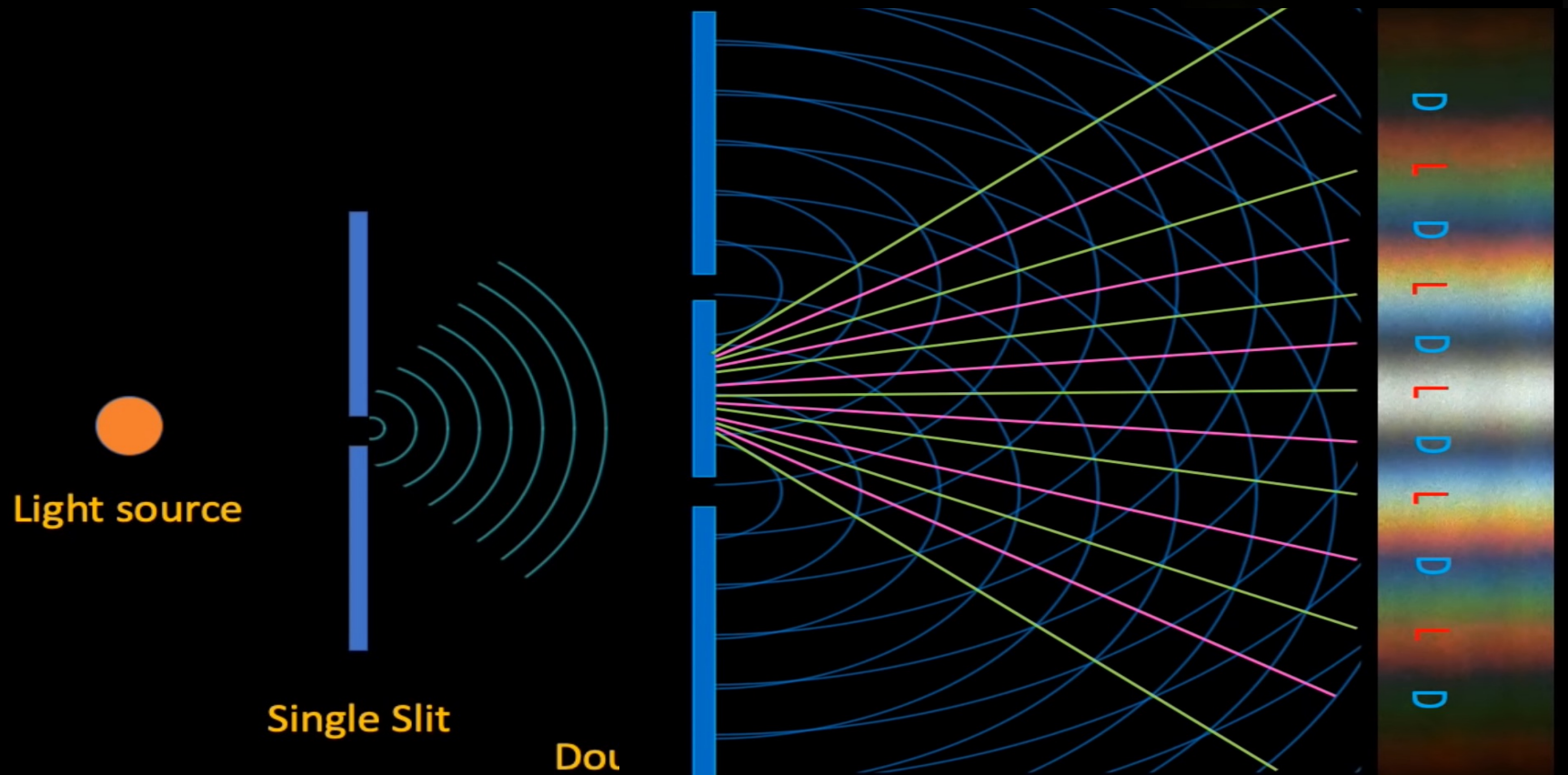


Thomas Young tried this more than 200 years ago and here is what he saw.

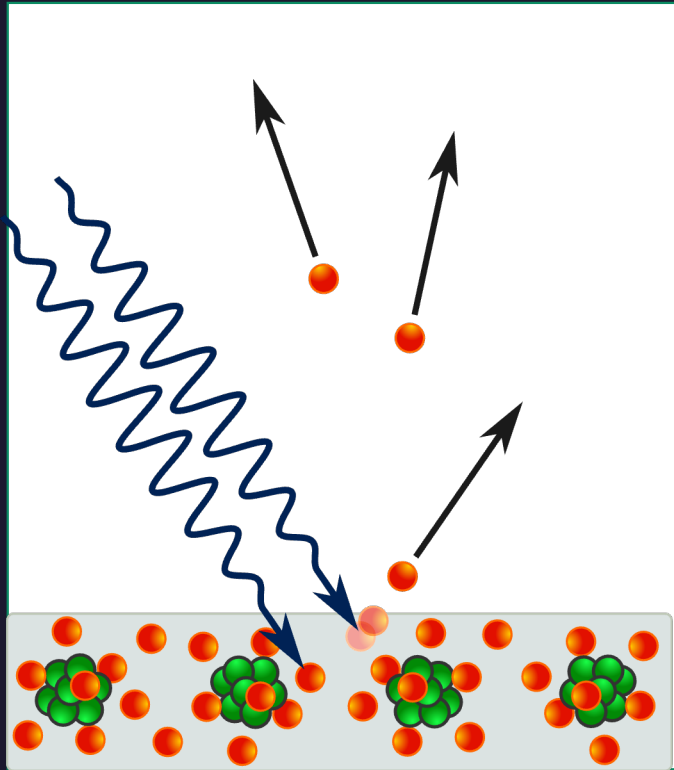
Why are
there more
than two slits
of light?



The bands are an interference pattern,
proving that light moves like a wave.



Light as a **particle**



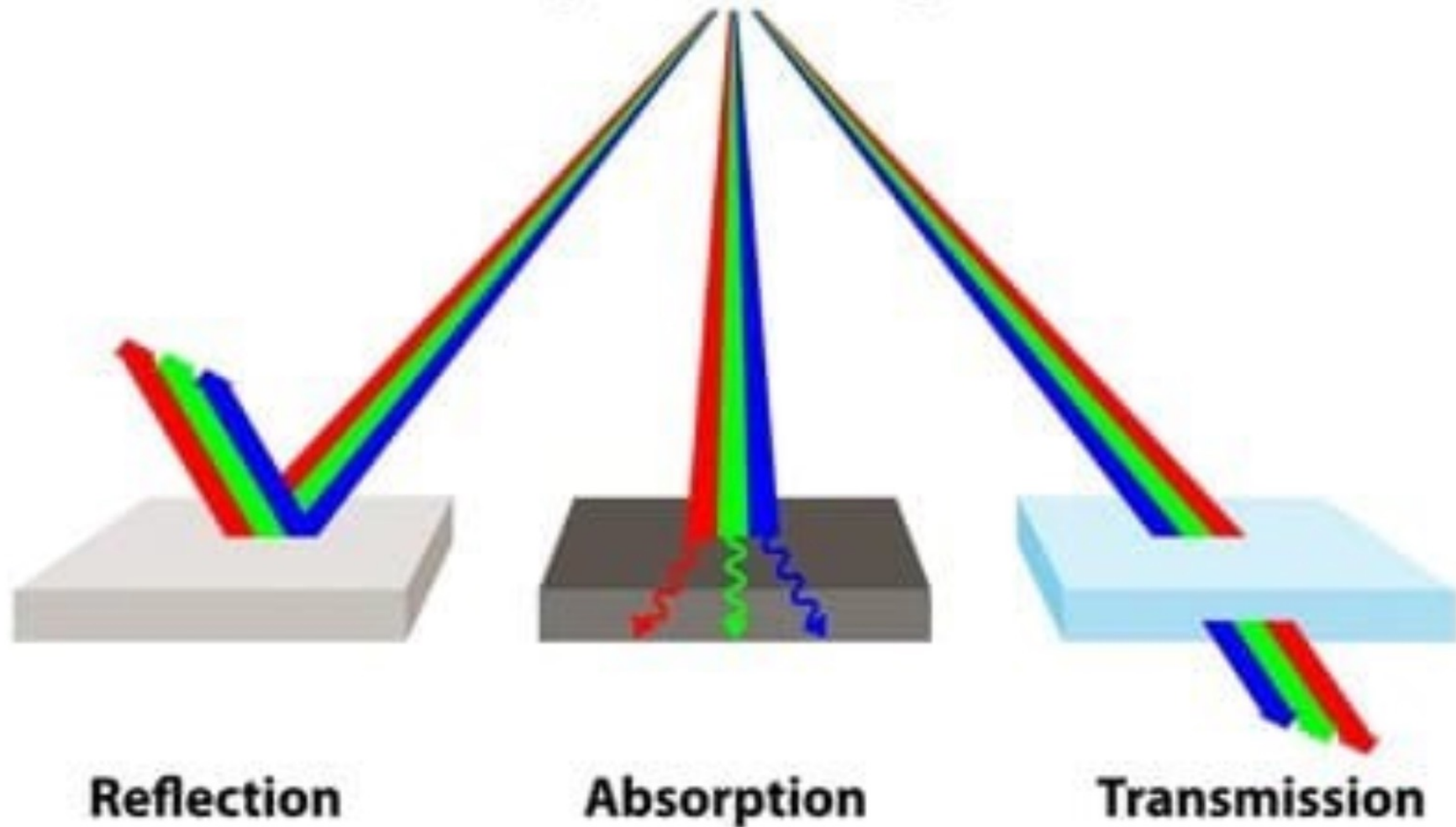
- One particle of light is called a **photon**
- A **photon** has energy, but no mass or electrical charge
- Physicists can show that light behaves like a particle by observing the **photoelectric effect**, which is when photons knock electrons free from the atoms in a material.

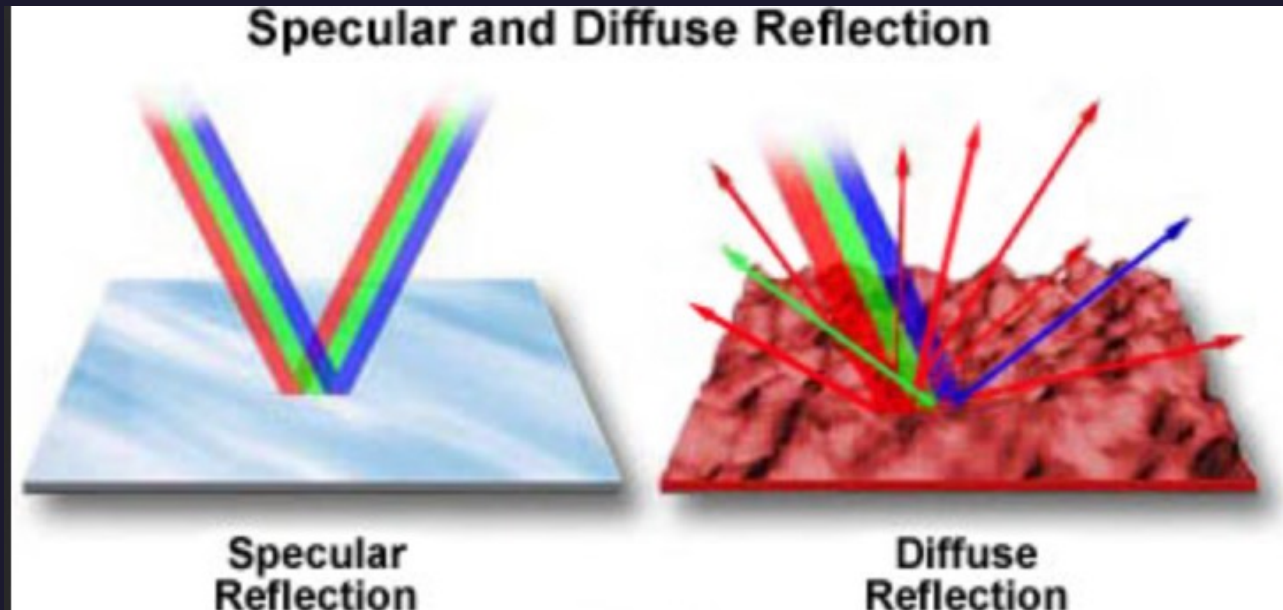


Since electromagnetic waves are not mechanical waves, they can move through a **vacuum**, a place with almost no matter.

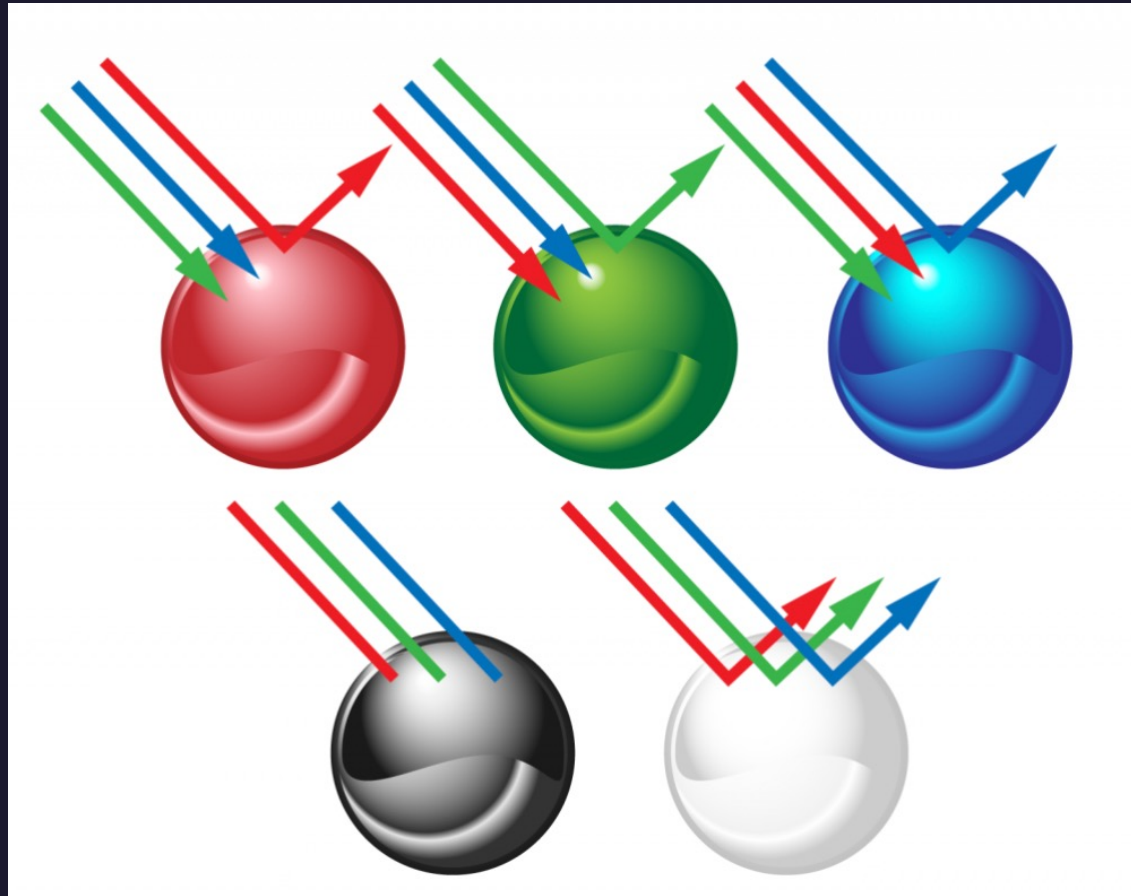
$$c = 299,792,458 \text{ m/s}$$

When light hits an object, it has 3 options

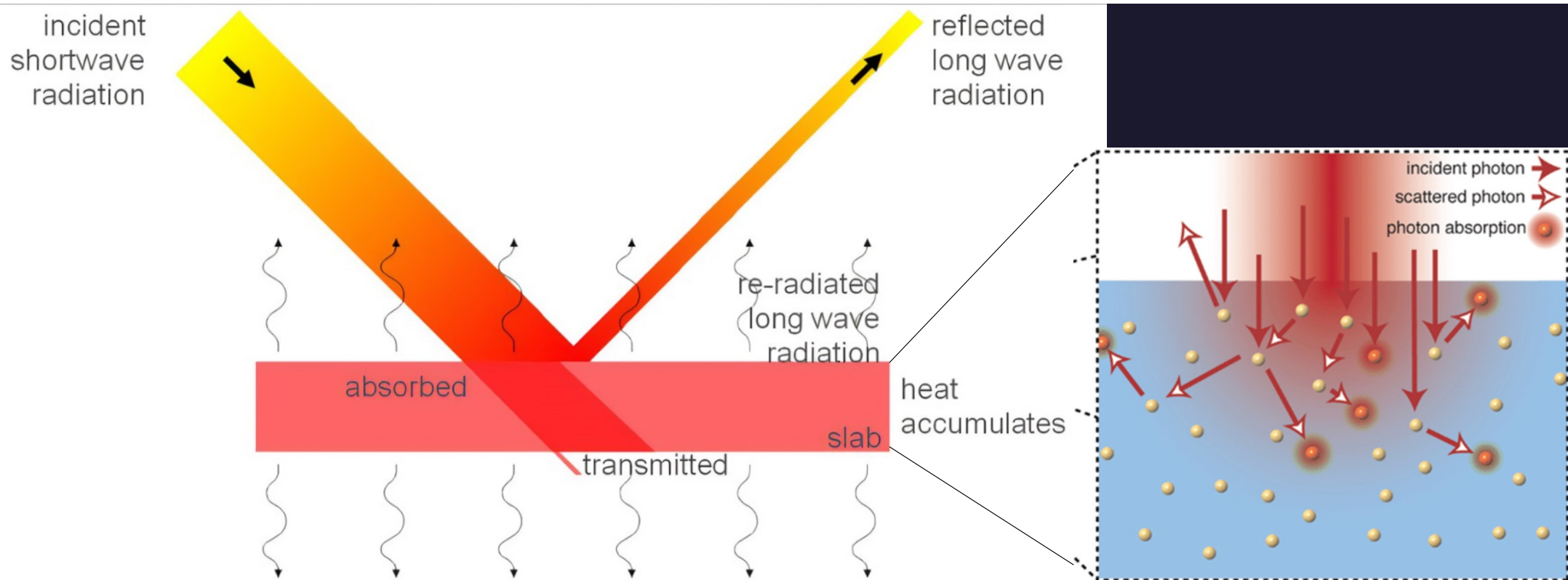




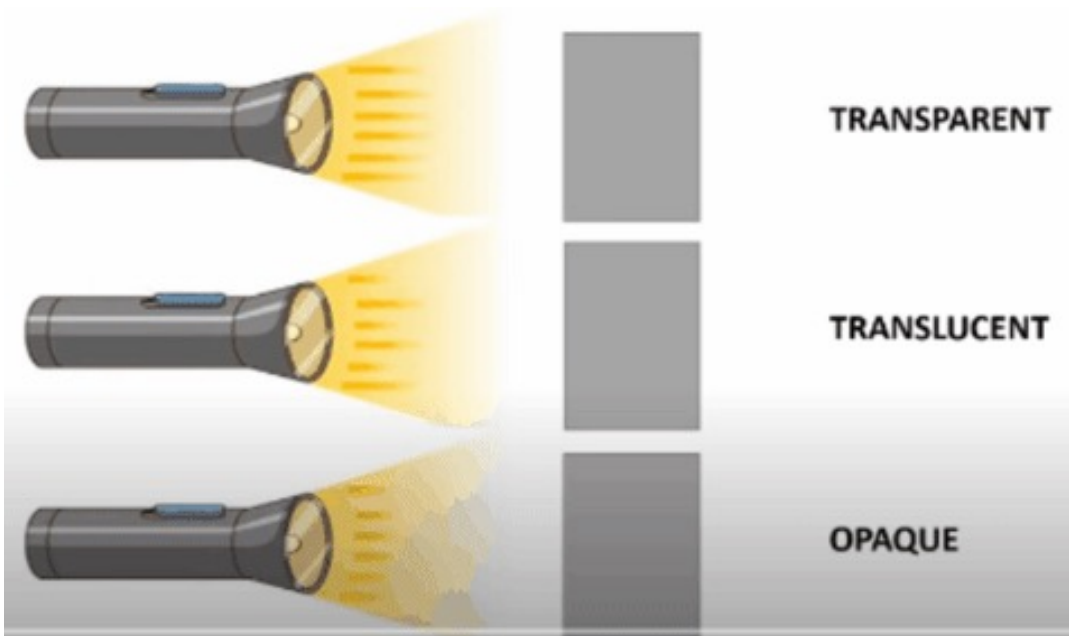
When light hits an object, some or all of it can be reflected. Reflected light may be diffused if the surface is not smooth.
Diffusion = scattered light



The colors we observe are the **reflected** wavelengths of light that have bounced into our eyes.

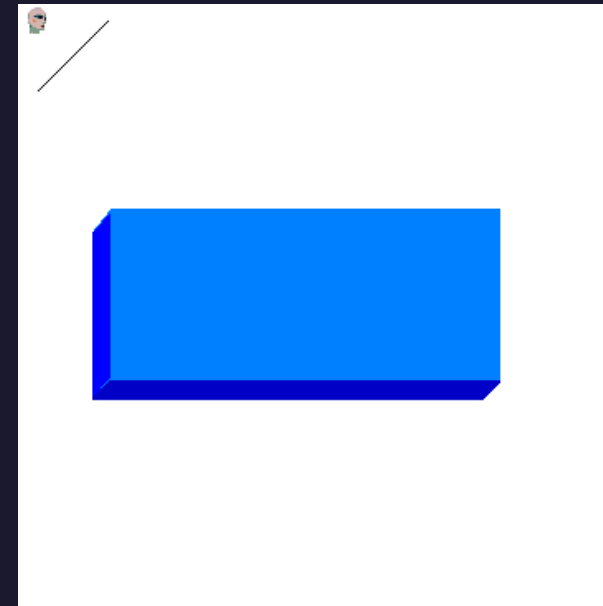
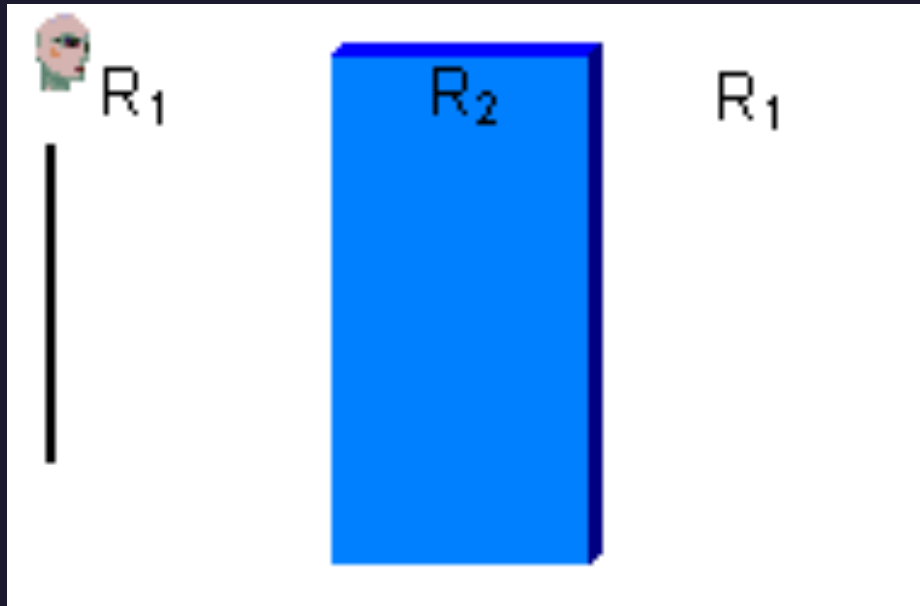


When light is **absorbed**, it may be transformed into **thermal energy**. The atoms of the object will be excited by the photons and vibrate faster.

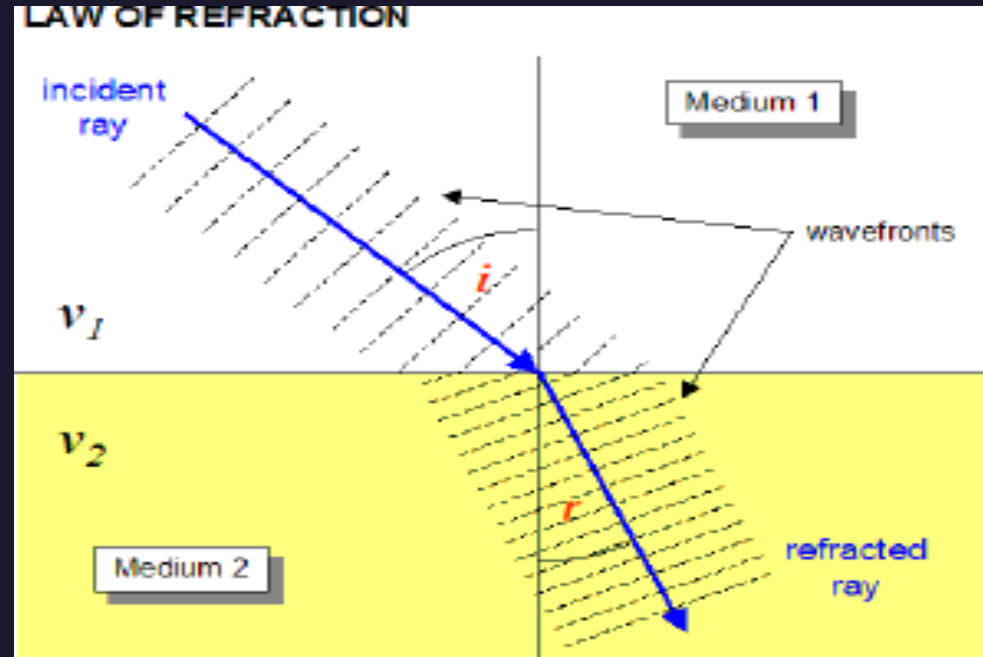


Transmission is when light passes through a medium (matter).

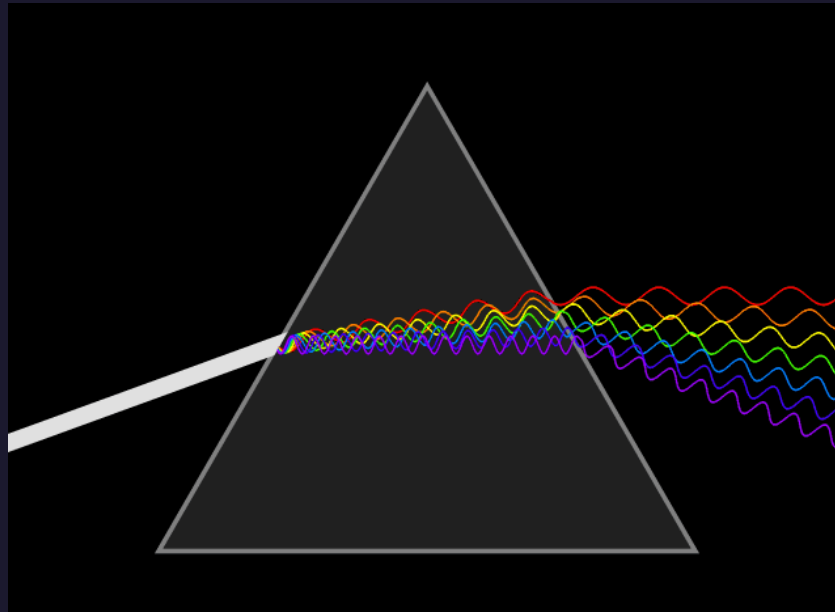
- A transparent medium is completely transmissible.
- A translucent medium is partially transmissible.
- An opaque medium is not transmissible.



When light is **transmitted**, it will change speed and direction. This is called **refraction**.
Denser **mediums** slow and bend the wave more.

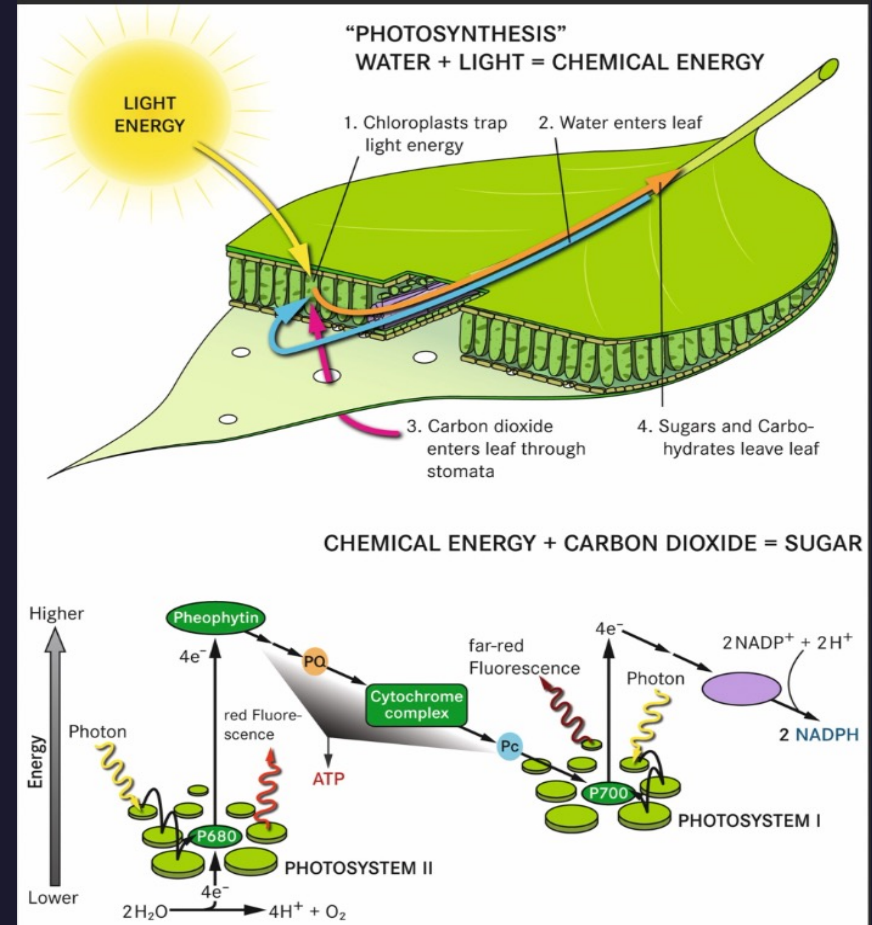
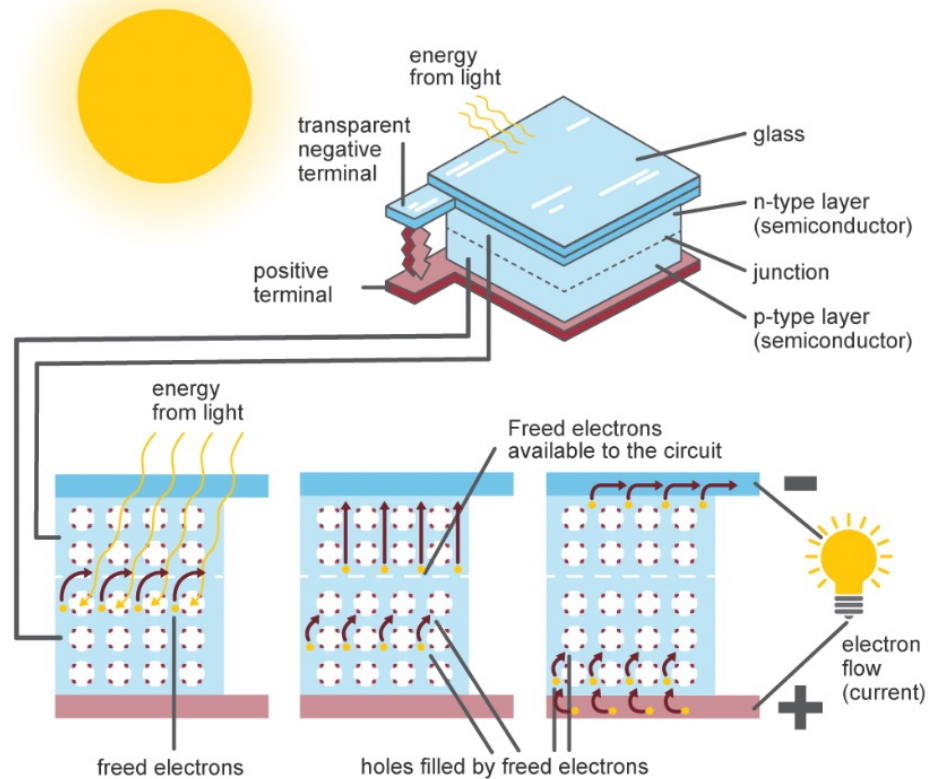


In Refraction, a medium changes the wavelength of transmitted light. Refracted light bends according to the density of the medium.



When visible light refracts, we see a rainbow because each color has a different angle of refraction. All the wavelengths added together appear as white light.

Inside a photovoltaic cell



What other forms of energy can radiant energy be transformed into?